

Nickel Hydroxide Formation in an Electrolytic Cell

Abstract

Ammonia electrooxidation can lower high concentrations of ammonia and make hydrogen transportation more feasible. Although ammonia electrooxidation currently requires expensive noble metal catalysts, there has been research indicating that nickel hydroxide catalysts can be created for the reaction. This article focuses on evaluating the most efficient methods for nickel hydroxide formation. A series of experiments show that nickel hydroxide is efficiently produced at a nickel cathode of an electrolytic cell, with a 0.085 M $\text{Ni}(\text{NO}_3)_2$ solution and a cathode current density of 1.1 mA/cm^2 .

Introduction

Ammonia causes water eutrophication at high concentrations, which fosters algae blooms that kill aquatic life. [8] Ammonia contamination can come from organic solvents, dyes, and pharmaceutical products that decompose into ammonia at wastewater plants. [8] Additionally, many industrial facilities release ammonia as by-product. Although permits are required for industries and wastewater plants to dispense substances into lakes, issues like flooding and inadequately treated discharges can lead to spikes in ammonia concentration. Non-point sources also heavily contribute to ammonia contamination since runoff from ammonia-based fertilizers can enter bodies of water. [1]

The most widely applied method to remove ammonia from water is bacterial degradation, but it is sensitive to pH, temperature, metals, and ammonia concentration. Air-stripping methods rely heavily on temperature, and breakpoint chlorination methods produce disinfection by-products. [7] On the other hand, ammonia electrolysis is cost-effective, clean, and easily implemented. The theoretical potential for ammonia oxidation is .077V, and it only produces hydrogen, nitrogen, and nitrates. Ammonia's ability to produce hydrogen is important because pure hydrogen has poor energy density by volume, considering the difficulty of transporting it. Hydrogen is easily combustible and must be cooled to a liquid for transportation or carried by gaseous tube trailers.

Meanwhile, ammonia is a liquid at room temperature, and there is already infrastructure around the globe to transport it. [5] Also, ammonia's energy density is similar to hydrogen by volume (11.38 GJm^{-3} for ammonia and 11.73 GJ m^{-3} for hydrogen). [3] Since ammonia produces hydrogen, ammonia can be a hydrogen carrier for fuel cells.

However, cheap and efficient catalysts are needed before the ammonia electrooxidation reaction can be applied. Ammonia electrooxidation works with platinum since its adsorption of nitrogen species follows the Sabatier principle. Often, iridium/platinum catalysts are used because iridium has a lower onset potential than platinum. Iridium and platinum are expensive, so chemists have attempted to make cheaper catalysts using combinations of nickel and other substances, like nickel hydroxide. [7]

In the ammonia electrooxidation reaction, nickel hydroxide turns into nickel oxyhydroxide. Nickel oxyhydroxide is the substance that oxidizes ammonia. The nickel hydroxide/nickel oxyhydroxide works best at pH levels above 7, and it does not deactivate as rapidly as platinum. Additionally, an increase in ammonia concentration up to 100mM leads to a continuous increase in current. [7] Conditions known to oxidize ammonia efficiently with the nickel hydroxide catalyst are as follows—electrolyte 0.01 M Na_2SO_4 , NH_3 concentration of 50 ppm, a current density of 1 mA/cm^2 , and pH 11 [8]. Ammonia electrooxidation with $\text{Ni}(\text{OH})_2$ begins with the dehydrogenation of adsorbed ammonia on the $\text{Ni}(\text{OH})_2$ and the reaction of two $\text{NH}_{x2}(\text{g})$. [8]

Materials and Methods

Preparation of solution and electrodes

Nickel (II) nitrate hexahydrate crystals from HiMedia were used to create a 0.085 M nickel nitrate solution. 0.99 grams of nickel (II) nitrate hexahydrate were measured using a top loading balance and placed into a 50-mL beaker. Then, distilled water was added to the beaker to a total of 40 mL. The solution was mixed for three minutes using a glass stirring rod.

Both the anode and cathode were made from a 0.22 mm thick pure nickel foam. Using scissors, a $2.30\text{ cm} \times 1.45\text{ cm}$ and a $2.50\text{ cm} \times 1.00\text{ cm}$ piece of the nickel foam was cut. The $2.30\text{ cm} \times 1.45\text{ cm}$ was the cathode, and the $2.50\text{ cm} \times 1.00\text{ cm}$ was the anode. Additionally, 16 cm of galvanized wire was cut for later use at the anode.

Preparation of the electrolytic cell

An H-type interchangeable membrane electrolytic tank was used as the container for the electrolytic cell. The lids at each side of the electrolytic cell had small holes so that wires could pass through. The galvanized wire was slid through one of the lids, so that each side of the lid had an equal length of wire. Using the galvanized wire stretching out from the bottom side of the lid, a small hole was pierced at one of the 1.00 cm edges of the anode. Once the hole was pierced, the galvanized wire was pushed through the anode by 1.00 cm, where it was bent upwards, away from the cathode. This way, the galvanized wire was secured to the anode. Then, the lid was then placed on the electrolytic tank so that the anode was suspended within the tank. Two pieces of electrical tape were used to secure the outside galvanized wire to the side of the tank. The nickel nitrate solution was poured into the tank until it reached 3 mm below the anode's hole. A black crocodile clip was slid through the other lid, and the clip was attached to the 1.45 cm edge of the cathode. The lid was placed on the electrolytic tank so that the cathode was suspended within the tank. The outside black crocodile clip was secured to the tank with electrical tape.

Running the reaction

A 1k potentiometer was amplified with a 470-ohm resistor so that it could control voltage on a 9V range. The circuit diagram is shown in figure 1. A fresh 9V battery was connected to the potentiometer. The other end of the black crocodile clip was connected to the negative end of the potentiometer. A red crocodile clip connected the positive end of the potentiometer to the red probe of a standard multimeter. The multimeter was set to 20m, and the multimeter's black probe was connected to the galvanized wire. Using the potentiometer, the current was adjusted to 3.19 mA for a current of 1.1 mA/cm^3 . Adjustments were made every 10 minutes to ensure that the

current reading on the multimeter was 3.19mA. The reaction ran for a total of 45 minutes. After 45 minutes, the cathode was taken out of the electrolytic cell and set to dry on a napkin for 3 hours.

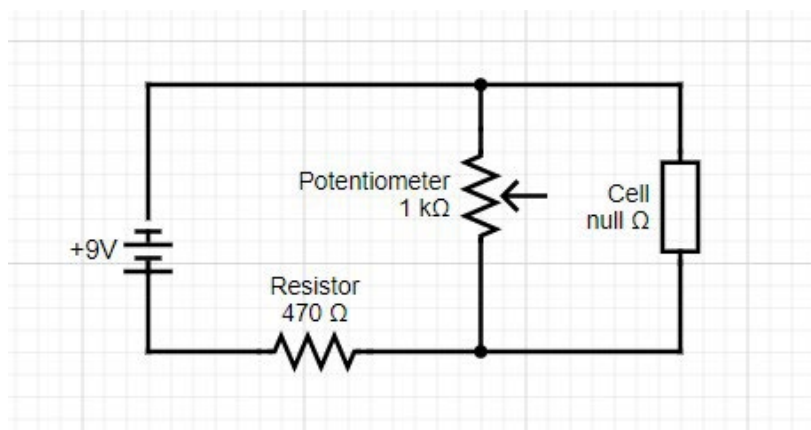


Figure 1

Results

Electrical Current Experiments Pictures of Cathode

The nickel hydroxide forms on the cathode. The pictures show a fully dried cathode from each experiment.



1.3 mA/cm²



1.1 mA/cm²



0.9 mA/cm²



0.7 mA/cm²



0.31 mA/cm²

Electrical Current; Current Experiment Tables

The current in each experiment had to be adjusted manually with a potentiometer to keep it somewhat constant. The initial current and voltage were measured before each adjustment. The adjusted current and voltage were measured after each adjustment. The 0.7 mA/cm² experiment data is not complete because the multimeter could not properly record the current after 30 minutes.

1.3mA/cm²

+	Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
	0	3.78	N/A	2.88	N/A
	10	4.09	3.78	2.80	2.69
	20	3.84	3.78	2.70	2.77
	30	3.00	3.77	2.72	3.35
	40	3.52	3.77	2.75	2.95
	50	3.75	N/A	2.85	N/A

1.1mA/cm²

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	3.21	N/A	2.95	N/A
10	3.54	3.18	2.85	2.68
20	3.30	3.22	2.65	2.70
30	3.32	3.25	2.60	2.55
40	3.80	3.22	2.51	2.54
50	2.78	N/A	2.55	N/A

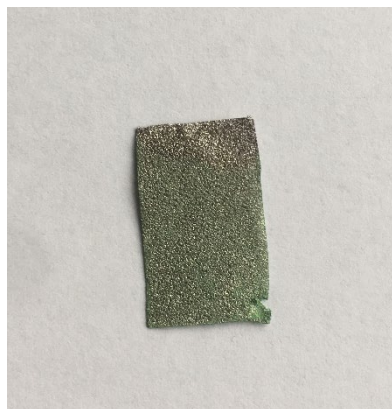
.9mA/cm²

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	2.50	N/A	3.34	N/A
10	2.29	2.65	3.20	2.79
20	2.77	2.65	2.68	2.75
30	2.65	2.65	2.79	2.75
40	2.25	2.66	2.77	3.00
50	2.87	N/A	3.06	N/A

.7 mA/cm²

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	2.13	N/A	2.70	N/A
10	2.18	2.08	2.23	2.25
20	2.14	2.10	2.25	2.30
30	-	2.10	2.30	2.90
40	-	2.10	2.35	2.70
50	-	N/A	2.65	N/A

Molarity Experiment Pictures of Cathode



0.16 M $\text{Ni}(\text{NO}_3)_2$



0.13 M $\text{Ni}(\text{NO}_3)_2$



0.085 M $\text{Ni}(\text{NO}_3)_2$



0.07 M $\text{Ni}(\text{NO}_3)_2$



0.04 M $\text{Ni}(\text{NO}_3)_2$

Electrical Current; Molarity Experiment Tables

0.16 M Ni(NO₃)₂

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	3.23	N/A	2.85	N/A
10	3.40	3.25	2.79	2.62
20	3.45	3.24	2.60	2.50
30	3.02	3.22	2.58	2.50
40	3.20	3.20	2.50	2.50
50	3.73	N/A	2.64	N/A

0.13 M Ni(NO₃)₂

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	3.18	N/A	2.98	N/A
10	3.61	3.18	2.85	2.70
20	3.41	3.23	2.70	3.00
30	3.96	3.24	2.68	2.55
40	3.75	3.20	2.83	2.80
50	3.52	N/A	2.79	N/A

0.085 M Ni(NO₃)₂

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	3.20	N/A	2.59	N/A
10	3.18	3.18	2.60	2.60
20	2.92	3.22	2.56	2.56
30	3.14	3.24	2.54	2.65
40	3.12	3.24	2.60	2.64
50	3.02	N/A	2.61	N/A

0.07 M Ni(NO₃)₂

Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	3.20	N/A	3.24	N/A
10	3.66	3.20	2.90	2.90
20	2.65	3.25	2.78	2.93
30	2.90	3.24	2.89	2.97
40	3.49	3.20	2.90	2.85
50	3.00	N/A	2.77	N/A

0.04 M Ni(NO₃)₂

+

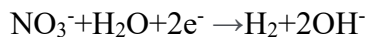
Time (minutes)	Initial current (mA)	Adjusted current (mA)	Initial voltage (V)	Adjusted voltage (V)
0	3.21	N/A	3.68	N/A
10	3.51	3.20	3.57	3.83
20	2.90	3.25	3.80	3.85
30	4.72	3.20	3.80	3.20
40	3.14	3.22	3.20	3.08
50	3.22	N/A	3.10	N/A

Discussion

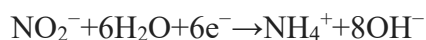
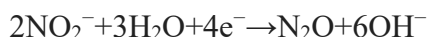
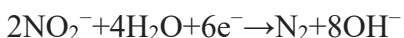
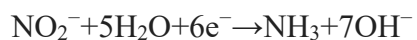
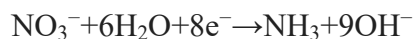
Nickel hydroxide forms due to OH⁻ produced at the cathode bonding with Ni²⁺ in the solution. Ni(OH)₂ produced is insoluble, so it precipitates onto the pores of the cathode. This means that only the formation of OH⁻ is electrochemical, and the formation of Ni(OH)₂ is a synthesis reaction. In nickel salt solutions, the OH⁻ is produced from the reduction of water, as given by the following equation. [6]



In $\text{Ni}(\text{NO}_3)_2$ solutions, though, the nitrate can produce hydroxide as well. [6]



There are more proposed reactions for hydroxide formation, which involve a further reduction of nitrite and the production of ammonium. These usually occur at high current densities (10 mA/cm² or more). [2]



The reaction at the anode depends on the type of anode material used. Nickel anodes oxidize to form Ni^{2+} and diffuse into the solution. Inert anodes oxidize water to form oxygen gas and hydroxyl ions. [2]



The anodes and cathodes used in the experiments for this article were all made of pure nickel foam. Since the reduction potential of nickel oxidation is less than water oxidation, the nickel anode should have oxidized. However, bubbling occurred at the anode throughout the entire 40 minutes of each experiment conducted, and there was no visible deterioration of the anode. If the

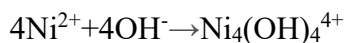
nickel oxidation reaction were to occur, no gas should not have been produced, and the nickel anode should have deteriorated to some degree. These two observations highly suggest that the main reaction occurring at the anode was water oxidation, which can be attributed to the voltage being greater than 1.23 V. The reaction at the cathode produced no visible bubbles, and its current density ranged from .3mA/cm² to 1.6 mA/cm², so it is highly likely that the reaction taking place was $\text{NO}_3^- + \text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$.

The water oxidation reaction is undesired since its reduction potential is 1.23 V [1] SHE. Meanwhile, nickel's reduction potential is -0.257 V SHE [2]. This means energy can be saved by solely oxidizing nickel rather than water. Furthermore, nickel oxidation, which produces Ni^{2+} , helps maintain the Ni^{2+} concentration in the solution. Meanwhile, the water oxidation reaction produces H^+ , which can interfere with the reaction between OH^- and Ni^{2+} at the cathode.

Still, initial nickel concentrations were high enough in the experiments conducted, so nickel hydroxide was produced. Additionally, the reaction only ran for 40 minutes, so the H^+ produced should not have interfered severely with the experiment results.

The experiment results show that as the current increases, the amount of nickel hydroxide deposited on the nickel cathode should increase. This remains true up to currents as high as 80 mA/cm². High currents result in the reduction of water rather than nitrate, which causes overheating of the catholyte. However, no experiments with currents that high were conducted for this article, so the decrease in nickel hydroxide precipitate formed beyond 80 mA/cm² is not shown. [6]

The molarity experiments showed efficient nickel hydroxide deposition from 0.04 M $\text{Ni}(\text{NO}_3)_2$ to 0.085 M $\text{Ni}(\text{NO}_3)_2$. Meanwhile, the 0.13 M $\text{Ni}(\text{NO}_3)_2$ and .16 M $\text{Ni}(\text{NO}_3)_2$ molarity experiments deposited very little nickel hydroxide. This is because at low $\text{Ni}(\text{NO}_3)_2$ concentrations, $\text{Ni}(\text{OH})_2$ is directly produced from the reaction between OH^- and Ni^{2+} , leading to faradaic efficiency of nearly 100%. Concentrations at 0.13M $\text{Ni}(\text{NO}_3)_2$ or higher formed $\text{Ni}(\text{OH})_2$ in the following two steps [2, 4].





The $\text{Ni}_4(\text{OH})_4^{4+}$ product from the first step is soluble in water and diffuses away from the substrate, so the second reaction is not likely to occur. $[\text{NiOH}]^+$ is another product that can be made in high concentrations, and just like $\text{Ni}_4(\text{OH})_4^{4+}$, it diffuses away [4]. This $\text{Ni}_4(\text{OH})_4^{4+}$ and $[\text{NiOH}]^+$ diffusion causes the low nickel hydroxide deposition efficiency in solutions above 0.13 M $\text{Ni}(\text{NO}_3)_2$.

In the conductivity and molarity experiments, two main types of nickel hydroxide could have been produced: $\beta\text{-Ni}(\text{OH})_2$ and $\alpha\text{-Ni}(\text{OH})_2$. $\beta\text{-Ni}(\text{OH})_2$ has a structure similar to $\text{Mg}(\text{OH})_2$. $\alpha\text{-Ni}(\text{OH})_2$ consists of layers of $\beta\text{-Ni}(\text{OH})_2$ intercalated by water molecules. These water molecules are free to move in between the layers and act as an amorphous glue that holds the layers together. Within the $\beta\text{-Ni}(\text{OH})_2$ and $\alpha\text{-Ni}(\text{OH})_2$ classifications, there are more differences based on structural disorder, like hydration, stacking fault disorder, ionic substitution, A/B interstratification, and internal mechanical stress. A more detailed explanation of each disorder type in nickel hydroxide is given by Baladev et al. Seemingly small distinctions can have an important effect on the properties of nickel hydroxide.

Because of a lack of materials, it was not possible to determine the structure of the nickel hydroxide catalysts created by the experiments detailed in this article.

References

1. "Ammonia." *United States Environmental Protection Agency*. 10 Dec., 2022.
<https://www.epa.gov/caddis-vol2/ammonia>
2. Ash Baladev, Venkata Swamy Nalajala, Ashok Kumar Popuri, Tondepu Subbaiah, and Manickam Minakshi. 2020. "Perspectives on Nickel Hydroxide Electrodes Suitable for Rechargeable Batteries: Electrolytic vs. Chemical Synthesis Routes." *Nanomaterials* 10, no. 9: 1878. <https://doi.org/10.3390/nano10091878>. Government
3. Barelli, Linda & Bidini, Gianni & Cinti, Giovanni. 2020. "Operation of a Solid Oxide Fuel Cell Based Power System with Ammonia as a Fuel: Experimental Test and System Design." *Energies*. 13. 6173. [10.3390/en13236173](https://doi.org/10.3390/en13236173)
4. Hall David S., Lockwood David J., Bock Christina and MacDougall Barry R. 2015. "Nickel Hydroxides and Related Materials: A Review of Their Structures, Synthesis and Properties." *Proc. R. Soc. A*. **471**2014079220140792.
<http://doi.org/10.1098/rspa.2014.0792>. Royal Society.
5. "Hydrogen Delivery." *Energy Efficiency and Renewable Energy*. 10 Dec., 2020.
<https://www.energy.gov/eere/fuelcells/hydrogen-delivery>
6. Kovalenko, Vadym & Kotok, Valerii & V., and Malishev. (2019). "Electrochemical Synthesis of Nickel Hydroxide From Salt in the Slit Diaphragm Electrolyzer for Power Sources Application." *Journal of Engineering and Applied Sciences*. 14. 2503-2510. Journal of Engineering.
7. Nadia Mohd Adli *et al* 2018 *J. Electrochem. Soc.* **165** J3130 DOI 10.1149/2.0191815jes
8. Yu-Jen Shih, Yao-Hui Huang , and C.P. Huang. 2018. "In-situ Electrochemical Formation of Nickel Oxyhydroxide (NiOOH) on Metallic Nickel Foam Electrode for the

Direct Oxidation of Ammonia in Aqueous Solution.” *The Official Journal of the ISE:*
Electrochimica Acta. 281. 410-419. <https://par.nsf.gov/servlets/purl/10091650>